

Table 1: Final controlled $B_{c1} \rightarrow B_c \gamma$ comparison. Widths are in keV. The present result is the leading hard-QCDSR value with the conservative G^2 systematic envelope obtained from the same-window heavy-heavy condensate screen. LHCb has observed the $B_c^+ \gamma$ structure, but the individual radiative partial widths are not measured.

Channel	Experiment	This work	Godfrey	Ebert et al.	Fulcher	Gershtein et al.	T. Y. Li et al.	Q. Li et al.	X. J. Li et al.	Eichten-Quigg	Bondar-Milstein
$B_{c1}(6743) \rightarrow B_c \gamma$	observed structure; partial width not measured	$0.223_{-0.061}^{+0.084} \pm 0.025_{G^2}$	13	18.4	32.4	11.6	16.6	35	30.1	9.9	22
$B_{c1}(6750) \rightarrow B_c \gamma$	observed structure; partial width not measured	$8.01_{-1.41}^{+1.89} \pm 0.91_{G^2}$	80	132	127.8	131.1	66.6	74	64	92.3	47

Table 2: Leading $B_{c1} \rightarrow B_c^* \gamma$ vector-baseline comparison. Widths are in keV. The present vector result uses the perturbative hard-photon three-point term and the physical f_1, f_2 residues. We show both the self-contained LO vector two-point normalization $f_{B_c^*} = 0.753_{-0.060}^{+0.055}$ GeV and the reference normalization $f_{B_c^*} = 0.384 \pm 0.038$ GeV, because the vector width is normalization dominated. Radiative G^2 and contact terms are not included.

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$B_{c1}(6743) \rightarrow B_c^* \gamma$	$24.8_{-3.0}^{+3.7}$ (LO $f_{B_c^*}$); $92.6_{-16.6}^{+24.4}$ (ref. $f_{B_c^*}$)	60	78.9	75.6	77.8	49	70	47.8	62.5	75
$B_{c1}(6750) \rightarrow B_c^* \gamma$	120_{-16}^{+17} (LO $f_{B_c^*}$); 447_{-83}^{+117} (ref. $f_{B_c^*}$)	11	13.6	26.2	8.1	10.5	40	25.6	7.5	2

Experimental status. The $B_c^+\gamma$ mass spectrum contains an observed $B_c(1P)^+$ -like structure, with a two-peak description around 6704.8 and 6752.4 MeV in the LHCb analysis. The partial radiative widths are not measured, so the experimental column should be read as an observation status rather than a numerical width comparison.

Interpretation. The $B_c\gamma$ channels are the present controlled prediction. The quoted first uncertainty is the Monte Carlo/input/Borel-window uncertainty of the leading perturbative hard-QCDSR calculation, and the second is the conservative G^2 systematic envelope. The $B_c^*\gamma$ channels are quoted as leading perturbative vector baselines. Their largest uncertainty is the normalization of $f_{B_c^*}$, so both the LO two-point and reference normalizations are displayed.